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Savitribai Phule Pune University, Pune Third Year of Artificial Intelligence and Machine Learning (2020 Course) 318556: Software Laboratory II		
Teaching Scheme:	**Credit Scheme:**	**Examination Scheme:**
Practical (PR) : 04 hrs./week	**1 Credit**	**PR : 25 Marks** **TW : 50 Marks**

Companion Course: 318552: Machine Intelligence for Data Science, 318554: Artificial Neural Network

Course Objectives :

- Students will demonstrate proficiency with statistical analysis of data.
- Students will execute statistical analyses with professional statistical software.
- Students will apply data science concepts and methods to solve problems.

Course Outcomes :

On completion of the course, students will be able to–

CO1: Demonstrate proficiency with statistical analysis of data.

CO2: Use statistical analyses with professional statistical software.

CO3: Apply data science concepts and methods to solve problems.

Guidelines for Instructor's Manual

Lab Assignments: Following is a list of suggested laboratory assignments for reference. Laboratory Instructors may design a suitable set of assignments for their respective courses at their level. Beyond curriculum assignments, the mini-project is also included as a part of laboratory work. The Inclusion of few optional assignments that are intricate and/or beyond the scope of curriculum will surely be the value addition for the students and it will satisfy the intellectuals within the group of the learners and will add to the perspective of the learners. For each laboratory assignment, it is essential for students to draw/write/generate flowchart, algorithm, test cases, mathematical model, Test data set and Comparative / complexity analysis (as applicable).

Guidelines for Student's Lab Journal

Program codes with sample output of all performed assignments are to be submitted as softcopy. Use of DVD or similar media containing students programs maintained by Laboratory In-charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory. As a conscious effort and little contribution towards Green IT and environment awareness, attaching printed papers as part of write-ups and program listing to journals may be avoided. Submission of journal/ term work in the form of softcopy is desirable and appreciated.

Guidelines for Lab /TW Assessment

Term work is continuous assessment that evaluates a student's progress throughout the semester. Term work assessment criteria specify the standards that must be met and the evidence that will be gathered to demonstrate the achievement of course outcomes. Categorical assessment criteria for the term work should establish unambiguous standards of achievement for each course outcome. They should describe what the learner is expected to perform in the laboratories or on the fields to show that the course outcomes have been achieved. It is recommended to conduct an internal monthly Oral examination as part of continuous assessment.

Guidelines for Laboratory Conduction

Following is a list of suggested laboratory assignments for reference. Laboratory Instructors may design a suitable set of assignments for respective courses at their level. Beyond curriculum assignments may be included as a part of laboratory work. The instructor may set multiple sets of assignments and distribute among batches of students. It is appreciated if the assignments are based on real world problems/applications. The Inclusion of few optional assignments that are intricate and/or beyond the scope of curriculum will surely be the value addition for the students and it will satisfy the intellectuals within the group of the learners and will add to the perspective of the learners. For each laboratory assignment, it is essential for students to draw/write/generate flowchart, algorithm, test cases, mathematical model, Test data set and comparative/complexity analysis (as applicable). Batch size for practical and tutorials may be as per guidelines of authority. **Use of open source software is to be encouraged.**

Guidelines for Practical Examination

Students' work will be evaluated typically based on the criteria like attentiveness, proficiency in execution of the task, regularity, punctuality, use of referencing, accuracy of language, use of supporting evidence in drawing conclusions, quality of critical thinking and similar performance measuring criteria.

List of Laboratory Assignments

Group A (Any 4) (Based on Machine intelligence for Data Science)

Assignment 1: Access an open source dataset "Titanic". Apply pre-processing techniques on the raw dataset.

Assignment 2: Text classification for Sentimental analysis using KNN. (Refer any dataset like Titanic, Twitter, etc.)

Assignment 3: Write a program to recognize a document is positive or negative based on polarity words using suitable classification method.

Assignment 4: Download Abalone dataset. (URL: http://archive.ics.uci.edu/ml/datasets/Abalone)

a) Predict the number of rings either as a continuous value or as a classification problem.
b) Predict the age of abalone from physical measurements using linear regression

Assignment 5: We have given a collection of 8 points.

P1=[0.1,0.6]
P2=[0.15,0.71]
P3=[0.08,0.9]
P4=[0.16, 0.85]
P5=[0.2,0.3]
P6=[0.25,0.5]
P7=[0.24,0.1]
P8=[0.3,0.2]

Perform the k-mean clustering with initial centroids as m1=P1 =Cluster#1=C1 and m2=P8=cluster#2=C2.

Answer the following

1] Which cluster does P6 belong to?

2] What is the population of cluster around m2?
3] What is updated value of m1 and m2?

Group B (Any 4) (Based on Artificial Neural Network))
Assignment 1: Write a program to scheme a few activation functions that are used in neural networks **Assignment 2:** Write a program to show back propagation network for XOR function with binary input and output **Assignment 3:** Write a program for producing back propagation feed forward network **Assignment 4:** Write a program to demonstrate ART **Assignment 5:**Write a program to demonstrate the perceptron learning law with its decision region using python. Give the output in graphical form